Characterizing HIV-1 Viral kinetics as a Pharmacodynamic Marker Using Nonlinear Mixed-Effects Modeling in ACTG 315

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Abstract

Understanding how quickly and how variably antiretroviral therapy (ART) suppresses HIV-1 is important for dose selection and regimen design for successful treatment. I developed a population nonlinear mixed-effects viral dynamics model using 361 HIV-1 RNA measurements from 46 patients in ACTG 315. A biphasic model, with baseline CD4 as a covariate on the second-phase rate, was fitted by maximum likelihood in "nlme" package and evaluated with diagnostic plots and a visual predictive check (VPC). Viral load started at about 5.1 log10 copies/mL. The first-phase decay rate was 0.31 per day (95% CI 0.27–0.35), corresponding to a half-life of 2.3 days, with low between-patient variability. The second phase was slow and highly variable between patients: its rate at median CD4 was close to zero. Higher baseline CD4 was associated with a faster second-phase decline (0.0105 per day per unit of centered CD4; 95% CI 0.004–0.017). Diagnostics showed no clear systematic bias, and the observed percentiles in the VPC remained within the simulated 90% intervals, indicating that the model reproduced both the typical trajectory and the between-patient variability. This is a pharmacodynamic model and cannot support dosing conclusions, but it provides a framework to which pharmacokinetics and exposure–response relationships can be added.

Background

HIV-1 is among the most serious infectious diseases worldwide, causing progressive immune deficiency and AIDS if untreated. There is still no cure for HIV. However, antiretroviral therapy (ART) can reduce the amount of virus in the blood (viral load) to a level so low that standard tests cannot detect it, and people who maintain an undetectable viral load do not transmit HIV through sex. Several classes of antiretroviral drugs are used to treat HIV, and the most widely used are integrase strand transfer inhibitors (INSTIs) and nucleoside reverse transcriptase inhibitors (NRTIs).

After antiretroviral therapy begins, plasma HIV-1 RNA falls in two phases: a rapid decline representing loss of short-lived productively infected cells and clearance of free virus, followed by a slower decline from longer-lived infected cell populations (Perelson et al, 1997). Because these rates summarize drug effect on the virus, they are used as pharmacodynamic markers. Nonlinear mixed-effects (NLME) modeling estimates the typical dynamics, the variability between patients, and the effect of covariates. Such a model describes how HIV behaves after treatment starts, and monitoring patients with it could help identify who is at risk of viral rebound, which can lead to failure to suppress viral load and potential transmission to others. Understanding what a drug does to the body, here what ART does to the virus, may therefore help us anticipate and ultimately, help prevent viral rebound and the failure of treatment.

Methods

Data

The ACTG 315 dataset included HIV-1 viral load measurements from 46 patients (361 observations) receiving protease inhibitors (PIs) and nucleoside reverse transcriptase inhibitors (NRTIs). Viral load was

analyzed on the $\log_{10}$ scale. Values at or below the assumed lower limit of quantification (LLOQ; 100 copies/mL) were replaced with $\log_{10}(50)$, corresponding to half the LLOQ. Baseline CD4 count was defined as each patient's first available CD4 measurement and was divided by 100 and centered around the population median for use as a covariate in the model.

Model. Viral load followed a biphasic decay:

$$DV(t) = \log10\{ \exp[p1 - \exp(ll1)\cdot t] + \exp[p2 - l2\cdot t] \} + \varepsilon$$

where p1 and p2 are the log-scale intercepts of the first and second phases, respectively, representing the baseline level of each phase. The first-phase decay rate is exp(ll1), and l2 is the second-phase decay rate. Baseline CD4 (centered at the median) was included as a covariate on l2. Random effects on all four parameters were assumed to be independent, and the residual error was additive on the log10 scale. The model was fitted by maximum likelihood using the "nlme" package (version 3.1.168) in R (Pinheiro & Bates, 2000).

Results

The biphasic decay model was fitted, and parameter estimates are presented in Table 1.

Table 1. Parameter estimates

Parameter	Estimate (SE)
p1	11.68 (0.20)
ll1 (log first-phase rate)	−1.19 (0.06)
p2	6.34 (0.25)
l2, intercept	−0.0005 (0.0032)
l2, CD4 effect	0.0105 (0.0034)
Between-patient SD	p1 1.11; ll1 0.26; p2 1.28; l2 0.017
Residual SD	0.38

The typical patient started at about 5.1 log10 copies/mL (roughly 120,000 copies/mL). In the first phase, viral load fell by about 26% per day (0.31 on the natural-log scale), a half-life of 2.3 days (95% CI 2.0–2.6). This phase varied little between patients. The second phase was slow: at median baseline CD4 its rate was close to zero, and the between-patient SD (0.017 per day) exceeded the mean, so some patients kept declining while others rebounded (Figure 1). Higher baseline CD4 was associated with a faster second-phase decline (0.0105 per day per unit of CD4; 95% CI 0.004–0.017).

Observed and individual predicted values agreed closely across 2–6 log10, apart from the band of imputed below-limit values, which is 11% of observations (Figure 2). The Pearson residuals (Figure 3) exhibited variable dispersion across the treatment period, with a wider spread at certain intermediate time points. Individual predictions followed each patient's observed profile closely, whereas population

predictions captured the average decline but not individual late-phase trajectories, including rebound in some patients (Figure 4). The VPC showed the observed 5th, 50th and 95th percentiles within their simulated 90% intervals throughout, so the model reproduced both the typical trajectory and the spread between patients. The observed median lay near the lower bound of its interval at late times, and intervals after day 50 were wide because of sparse sampling (Figure 5).

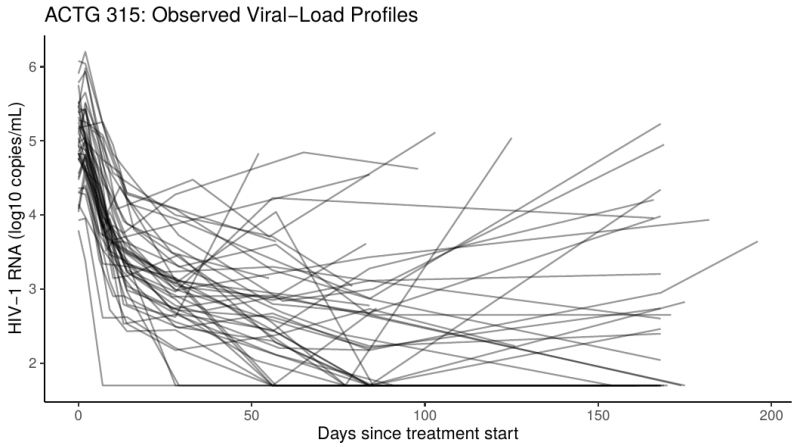


Figure 1. Observed HIV-1 RNA profiles in ACTG 315.

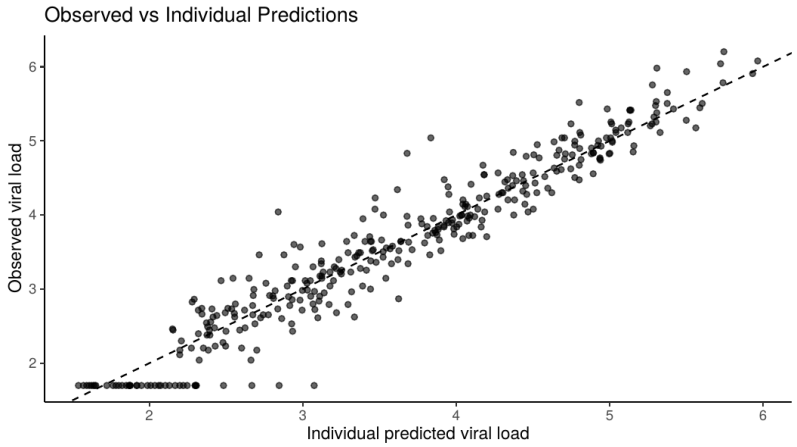


Figure 2. Observed vs individual predicted viral load (dashed line: identity). The horizontal band is imputed below-limit values (censored data).

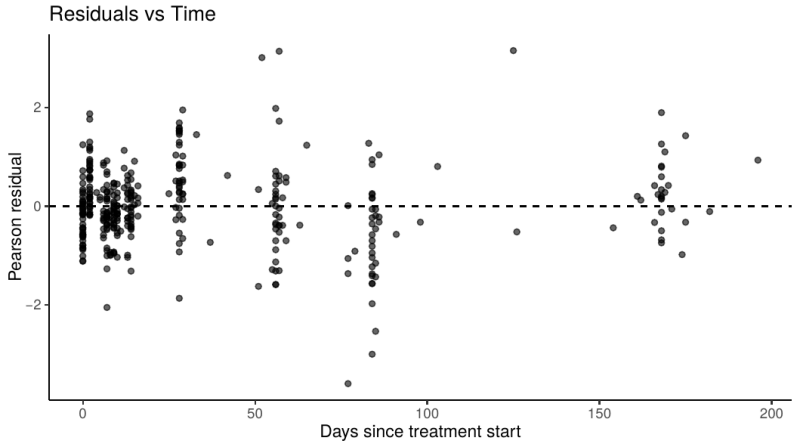


Figure 3. Pearson residuals vs time.

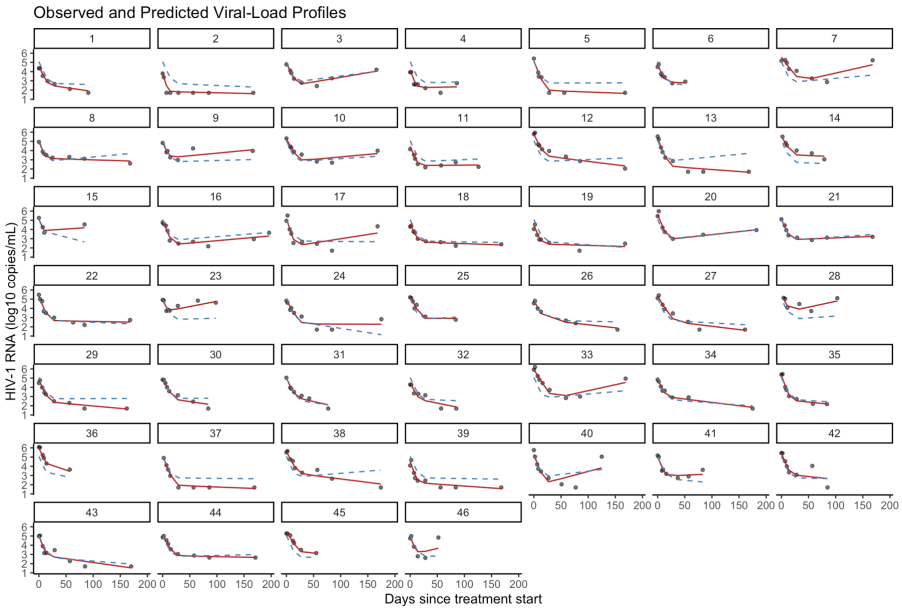


Figure 4. Observed (point), individual predicted (red) and population predicted (blue) viral load for each of the 46 patients.

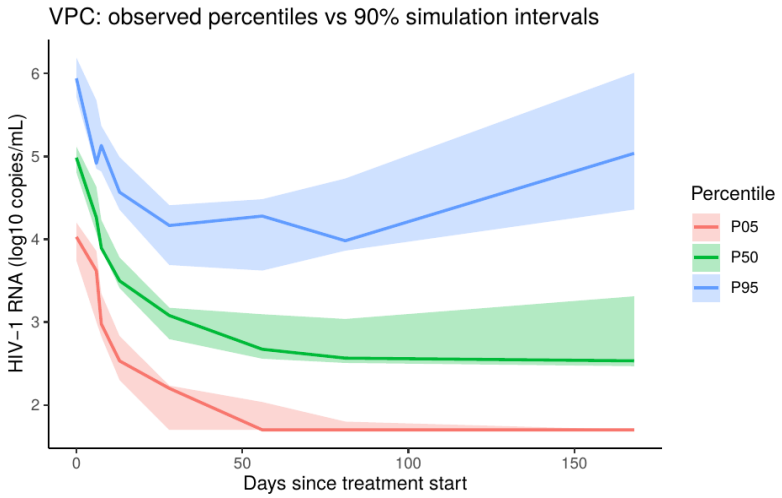


Figure 5. Visual predictive check: observed percentiles (lines) vs 90% simulation intervals (bands).

Discussion and future direction

Summary of findings. A biphasic nonlinear mixed-effects model with baseline CD4 as a covariate described HIV-1 RNA dynamics in 46 patients from ACTG 315. The pattern matches the established view of HIV-1 dynamics under therapy: a rapid first-phase decline, consistent with loss of short-lived infected cells, followed by a slower second phase from longer-lived populations. The first-phase half-life of 2.3 days implies an average lifespan of about 3 days for short-lived infected cells, assuming near-complete drug effect. The narrow between-patient spread in this phase suggests a similar early antiviral effect across patients. The much wider spread in the second phase indicates that longer-term outcomes differ markedly.

Model performance. Observed and individual predicted values aligned closely, and Pearson residuals were centered on zero, though their dispersion was uneven over time (Figure 3). This pattern may represent interindividual heterogeneity in HIV-1 viral-load trajectories during treatment. Patients may differ in the rate of viral decay, the persistence of virus-producing cells, and the dynamics of viral suppression or rebound. These differences may not be fully captured by the current biphasic viral dynamics model. Individual predictions followed each patient's profile, whereas population predictions captured only the average decline and missed late rebound in some patients (Figure 4). The visual predictive check, a stronger test because it compares simulated with observed data, showed the observed 5th, 50th and 95th percentiles within their 90% simulation intervals throughout (Figure 5). The model therefore reproduced both the typical trajectory and the variability between patients. The wide intervals after day 50 reflect sparse sampling and limit our power to detect misfit there. The uneven residual spread and rebound in some patients indicate that between-patient differences in decay rate, persistence of virus-producing cells and timing of rebound are not fully captured. Possible causes of rebound include resistance, imperfect adherence and low drug exposure, which these data cannot separate. Higher baseline CD4 was associated with a faster second-phase decline, which may reflect better immune clearance or a smaller long-lived reservoir.

Relevance to drug development. Although the data are retrospective and contain no drug concentrations, the framework is the one used in model-informed drug development, where models support dose selection, trial design and regulatory decisions. Models of this type contribute to three

visions: **1) Quantifying drug effect early.** The first-phase decay rate summarizes how fast therapy reduces viral load and varied little between patients. As a short-term pharmacodynamic marker, it allows regimens to be compared within days to weeks, before long-term outcomes are available. The model also separates early drug effect (first phase) from long-term control (second phase), which a single time-point measurement cannot do. **2) Identifying covariates that explain response.** Baseline CD4 was associated with second-phase decline. In a drug development program, a covariate effect of this kind can suggest which subgroups need closer monitoring or a different regimen, and which baseline measures should be used to stratify randomization. Pharmacogenomic studies of patients who are likely to rebound could also help identify the mechanisms underlying loss of drug efficacy.
3) Linking exposure to response. Adding a pharmacokinetic model and an Emax or inhibition term linking concentration to viral production or infectivity would turn this model into a full PK/PD model. It could then be used to simulate alternative doses, missed doses and dosing intervals, and to ask how much exposure keeps viral load suppressed.

Limitations. Below-limit values were substituted rather than modeled as censored data, which compresses the late decline and may bias second-phase estimates. The second-phase intercept is weakly identified, and all mechanistic interpretations should be regarded as hypotheses.

References

- Perelson AS, Essunger P, Cao Y, Vesanen M, Hurley A, Markowitz M, et al. Decay characteristics of HIV-1-infected compartments during combination therapy. Nature. 1997;387(6629):188–191. doi: 10.1038/387188a0
- Lederman MM, Connick E, Landay A, et al. (1998), "Immunologic responses associated with 12 weeks of combination antiretroviral therapy consisting of zidovudine, lamivudine and ritonavir: results of AIDS Clinical Trials Group Protocol 315," Journal of Infectious Diseases, 178: 70-79.

Supplementary

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Summary(fit)

Nonlinear mixed-effects model fit by maximum likelihood
  Model: DV ~ log10(exp(p1 - exp(ll1) * day) + exp(p2 - l2 * day))
  Data: dat
       AIC      BIC    logLik
  622.9183 661.8071 -301.4592

Random effects:
 Formula: list(p1 ~ 1, ll1 ~ 1, p2 ~ 1, l2 ~ 1)
 Level: patid
 Structure: Diagonal
             p1       ll1       p2 l2.(Intercept)  Residual
StdDev: 1.113786 0.2638222 1.277331      0.0170109 0.3816305

Fixed effects:  list(p1 ~ 1, ll1 ~ 1, p2 ~ 1, l2 ~ CD4B)
                    Value  Std.Error  DF   t-value p-value
p1              11.676093 0.19604554 311  59.55806  0.0000
ll1             -1.187155 0.06318803 311 -18.78765  0.0000
p2               6.335727 0.24545691 311  25.81197  0.0000
l2.(Intercept) -0.000537 0.00323569 311  -0.16605  0.8682
l2.CD4B         0.010532 0.00343603 311   3.06528  0.0024
 Correlation:
                p1    ll1   p2    l2.(I)
ll1             0.310
p2              0.030 0.140
l2.(Intercept) 0.022 0.108 0.337
l2.CD4B        0.004 0.022 0.052 0.114

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-3.60461909 -0.43859022 -0.02548713  0.41842721  3.16144410

Number of Observations: 361
Number of Groups: 46
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